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Grade	**0** out of 100

Question **1**

Not answered

Marked out of 1

Two hospitals in Nottinghamshire are managed by the same NHS trust. The managers note that the duration of hospital admission in the inpatient units to be significantly different between them affecting the bed occupancy rates unfavourably in one area. If the managers intend to compare the length of stay in these two hospitals, which of the following methods is preferable?

Select one:

- Chi square test of 'long' admissions in each hospital
- Survival methods measuring 'time to discharge'
- Analysis of variance of admission duration
- 'T' test of mean admission duration
- F test of mean admission duration

Your answer is incorrect.

Survival methods are best suited to measuring "time to an event" i.e. discharge from hospital. These methods do not require a near-normal distribution of values; such parametric assumptions are needed for a t-test or F-test (ANOVA). The chi-square test, in this case, would involve loss of significant information due to dichotomisation of the data.

The correct answer is: Survival methods measuring 'time to discharge'

Question 2

Not answered

Marked out of 1

In survival analysis using Kaplan-Meier method, which of the following is plotted on the y axis?

Select one:

- Time remaining after achieving an outcome
- Proportion of those who achieved the outcome of interest
- Proportion of those who have not achieved the outcome of interest
- Life expectancy at each point of observation
- Time taken for achieving an outcome

Your answer is incorrect.

In analyzing survival data, two functions that are dependent on time are of particular interest: the survival function and the hazard function. The survival function S(t) is defined as the probability of surviving at least to time t. The hazard function h(t) is the conditional probability of dying at time t having survived to that time. The graph of S(t) against t is called the survival curve (Retrieved from http://www.ncbi.nlm.nih.gov/pmc/articles/PMC1065034/). The Kaplan-Meier method can be used to estimate this curve wherein S(t) is given by proportion of individuals surviving an anticipated outcome - this is plotted along the y-axis while time period is plotted across the x-axis. Note that survival curves are sometimes plotted with a logarithmic vertical scale, making the survival curves look like straight lines. This may be misleading if one does not attend to the plots expressed.

The correct answer is: Proportion of those who have not achieved the outcome of interest

Question 3

Not answered

Marked out of 1

Which of the following is the most important outcome of interest in survival analysis?

Select one:

- Cost of setting up a service
- Number of cases surviving an intervention
- Time taken to achieve an event of interest
- Proportion responding to a treatment
- Mean score achieved by two groups at the outcome

Your answer is incorrect.

The probability of surviving a condition for a given duration is the most important parameter of interest for survival curves. This can also yield time taken to achieve an event of interest in individuals belonging to the observed cohort.

The correct answer is: Time taken to achieve an event of interest

Question 4

Not answered

Marked out of 1

Which of the following tests can be applied if you want to test if smokers differ significantly from non-smokers with respect to risk of depression over the 10 year period, irrespective of other systematic differences between the two groups?

Select one:

- T statistics
- Chi square test
- Log rank test
- Mann Whitney U test
- Cox proportional regression test

Your answer is incorrect.

The log-rank test enables us to assess the influence of a categorical variable on survival.

The correct answer is: Log rank test

Question 5

Not answered

Marked out of 1

Which of the following tests can be applied if you want to test if smokers differ significantly from non-smokers with respect to risk of depression over the 10 year period after adjusting for the confounding effect of age and alcohol intake?

Select one:

- Log rank test
- Mann Whitney U test
- T statistics
- Cox proportional regression test
- Chi square test

Your answer is incorrect.

The log-rank test enables us to assess the influence of a categorical variable on survival. But if we want to assess the influence of a continuous variable, such as age, weight, or blood pressure, we need proportional hazards (or Cox) regression (Excerpt retrieved from https://www.med.illinois.edu/m2/epidemiology/tahandouts/survival). This method allows us to consider several variables simultaneously. Thus, we can produce estimates of the effect of each variable, adjusting for the others. This can help us distinguish between a variable that has an independent effect on survival, and one whose effect on survival is due only to its association with another variable

The correct answer is: Cox proportional regression test

Question 6

Not answered

Marked out of 1

Which of the following assumptions must be met before applying Cox's proportional hazard test?

Select one:

- The survival curves cross each other during the period of follow-up
- The median survival time is reached during the period of follow-up
- The hazard ratio is constant over time
- At least 50% of the observed events must have occurred in the first year of observation
- Confounders have no effect on survival time

Your answer is incorrect.

We can use either log-rank test or Cox test to determine statistically the difference between two survival curves. The log-rank test is used to test whether there is a difference between the survival times of different groups, but it does not allow other explanatory variables to be taken into account. Cox's proportional hazards model is analogous to a multiple regression models and enables the difference between survival times of particular groups of patients to be tested while allowing for other factors. In this model, the response (dependent) variable is the 'hazard'. The hazard is the probability of dying (or experiencing the event in question) given that patients have survived up to a given point in time, or the risk for death at that moment. In Cox's model no assumption is made about the probability distribution of the hazard (Retrieved from http://www.ncbi.nlm.nih.gov/pmc/articles/PMC1065034/). However, it is assumed that the ratio of the risk for dying at a particular point in time between one group and the other is constant all the time. In other words, Cox test assumes that the hazard ratio does not depend on time.

The correct answer is: The hazard ratio is constant over time